Nicolò Iacobone

Computer Vision & 3D Deep Learning Researcher

Bari, Italy · open to relocation worldwide · EU citizen · nicolo.iacobone@gmail.com · +39 391 370 2668
linkedin.com/in/nicoloiacobone · github.com/NicoloIacobone · nicoloiacobone.it

About Me

I am a computer vision researcher working across feed-forward 3D reconstruction and semantic scene understanding. I am looking for research or ML engineering roles in industry, or a PhD in 3D perception and representation learning. Outside engineering, I build things with Arduino, I juggle, fish and solve Rubik's cubes, and every season I reach Grand Champion in Clash Royale, which puts me among the top 2–4% of active players worldwide.

Education

MSc in Computer Engineering — Artificial Intelligence and Data Analytics **Oct 2022 – Mar 2026**
Politecnico di Torino, Italy · 108/110 · Thesis carried out at ETH Zürich (Jul 2025 – Mar 2026)

BSc in Computer Science and Automation Engineering **Oct 2019 – Oct 2022**
Politecnico di Bari, Italy · Erasmus exchange at Universitatea Tehnică din Cluj-Napoca, Romania (Oct 2021 – Jul 2022)

Research Experience

Research Collaborator — ETH Zürich, Photogrammetry & Remote Sensing (D-BAUG) **Mar 2026 – Sep 2026**
3D-Consistent Instance Segmentation on Frozen Feed-Forward Backbones · *Code*

- Taught a frozen 3D reconstruction model (VGGT) to also pick out the individual objects in a scene, keeping each object consistent across every view, while training only on ordinary 2D annotations.
- Built the whole pipeline myself, from data loading and backbone feature caching to training on a SLURM cluster, evaluation on the official ScanNet benchmark and ablation studies.

Master's Thesis — ETH Zürich **Jul 2025 – Mar 2026**
Towards Multi-View 3D Instance Segmentation via Semantic Distillation · *Code* · *SAM 2 teacher fork*

- Taught a 3D reconstruction model (MapAnything) to also learn semantic features by knowledge distillation, imitating those of a segmentation model (SAM 2) while staying consistent across views.
- Found that a geometry-focused model runs out of capacity for fine-grained semantics, which led to the follow-up work above.

Projects

Independent work on open-source models

- **Automatic 2D instance annotation of ScanNet with SAM 3** · code **May – Jun 2026**
  A pipeline that labels objects from a fixed 19-class list across ScanNet scenes, tracking them across frames and picking them up again when lost.
- **Controlled synthetic 3D data and point tracking** · Kubric · SpaTrackerV2 **Sep 2025**
  Generated synthetic scene configurations with configurable camera motion and ground-truth object masks, from still objects to several moving ones, and ran batch 3D point tracking on them.

Personal projects

- **Doppelbot:** **Ongoing**
  ROS 2 digital twin, a Unitree G1 in RViz that mirrors my body and hands via MediaPipe
- **OpenFactory AI:** **Ongoing**
  building a ROS 2 bin-picking cell in simulation: UR5e, Gazebo, MoveIt 2, ros2_control
- **Rubik's cube scanner & solver in the browser** · live demo **Oct 2026**
  Scans the cube with OpenCV.js and guides the solution, entirely in the browser.

Academic projects

- **Real-Time Anomaly Detection for Road Scenes** · Report **Jun 2024**
  Real-time inference on embedded devices
- **Synthetic Data Generation for AROL Closure Systems** · Code & slides **Mar 2024**
  CVAE for sensor time series, developed with an industrial partner
- **HoloLens 2 Mixed-Reality App** · Code **Jun 2022**
  Unity/MRTK holograms that follow the user's head and hands

Teaching & outreach

- **Arduino orientation course, Politecnico di Torino** · Tiny Sorter **Feb – Jun 2024**
  Introductory course for final-year high-school students, ending with a challenge to build a two-color M&M's sorter with a servo and the computer's camera.

Skills

3D Computer Vision: 3D reconstruction & scene understanding · multi-view perception · 2D/3D instance segmentation · 3D point tracking

Deep Learning: Knowledge distillation · object detection & segmentation · representation learning · synthetic data generation

Robotics: ROS 2 · MoveIt 2 · Gazebo

Programming: Python (advanced), C/C++, PyTorch, HPC / SLURM **Languages:** Italian (native) · English (proficient)